Acoustic Source Localization with Unknown Attenuation using the Grey Wolf Optimizer

Swarm Intelligence mini-project · Grey Wolf Optimizer (continuous encoding) · primary seed 4612 · code: `gwo_acoustic.ipynb`

1. Introduction

A sound source somewhere in a 50 × 50 × 10 m hall is recorded by a microphone array. From the sound-pressure levels alone we want its 3-D position, its source level *S* and the propagation attenuation α — five continuous unknowns, coupled non-linearly through distance and corrupted by measurement noise. The misfit landscape is smooth but contains a long, flat, curved valley (S, α and range can partly compensate each other) and an obstacle the source cannot be inside. We solve it with a from-scratch Grey Wolf Optimizer (GWO), compare it with random search at an identical budget, study its parameters, stress it with more noise, fewer microphones, impulsive *shot noise* and different source locations, ask which parameter drives the objective most, and ablate its three-leader guidance. Every number below is reproducible from seed 4612.

2. Problem formulation

Instance (seed 4612). One generator `default_rng(4612)` draws, in order: the obstacle (edges U[4–12, 4–12, 3–7] m, standing on the floor) → the true source (rejection-sampled 1–4 m outside the box) → S ~ U(85,105) dB → α ~ U(0.05,0.30) dB/m → 20 microphones uniform in the hall (outside the box) → noise shape z_i ~ N(0,1) → the 4 microphones removed in the stress test. Result: box [30.0, 37.9] × [22.9, 34.4] × [0, 5.7] m; source s = (29.75, 21.85, 1.11) m, 1.09 m from the box; S = 93.47 dB; α = 0.0842 dB/m; removed microphones {0, 14, 18, 19}. Noise at level σ is $\varepsilon_i = \sigma z_i$.

Variables x = [s_x, s_y, s_z, S, α]. **Model** and **objective**:

$$L_i^{\text{pred}}(x) = S - 20\cdot\log_{10} r_i - \alpha\cdot r_i, \quad r_i = \max(\|m_i - s\|, 0.1\ \text{m}), \quad f(x) = (1/N)\ \Sigma_i\ (L_i^{\text{meas}} - L_i^{\text{pred}}(x))^2 \ \ [\text{dB}^2]$$

Under Gaussian noise the minimiser of f is the maximum-likelihood (ML) estimate and $f(x_{\text{true}}) \approx \sigma^2$. **Constraints:** (a) bounds x,y ∈ [0,50] m, z ∈ [0,10] m, S ∈ [60,120] dB; (b) $0 \le \alpha \le \alpha_{\max} = 0.5$ dB/m; (c) source outside the obstacle box B.

$$F(x) = f(x) + \lambda_{\text{obs}}\cdot d_B(s)^2 + \lambda_\alpha\cdot[\max(0, -\alpha) + \max(0, \alpha - \alpha_{\max})]^2$$

Constraint handling. (a) by *reflection* at the violated bound after every move (then clipping), which avoids wolves piling up on the boundary. The α encoding box is [−0.05, 0.55], wider than the physical range, so (b) and (c) are penalised. d_B is the *penetration depth*: distance to the nearest escapable face (0 outside; the floor face is not escapable, and a point on the floor under the box is inside). **λ choice:** $\lambda_{\text{obs}} = 10^2$ dB²/m², $\lambda_\alpha = 10^4$ dB²/(dB/m)² make a violation of one resolution unit (0.1 m, 0.01 dB/m) cost ≈ σ^2 = 1 dB² — the noise floor — whereas 1 m inside the box costs 100 dB², more than nearly any feasible misfit. **Why graded, not a death penalty:** a constant/infinite penalty creates a flat plateau over the obstacle that gives the leaders no direction back to feasibility; the squared depth is zero with zero slope on the surface (feasible boundary solutions are undistorted) and grows smoothly inside, so the pack can cross the obstacle region while exploring and is pushed out proportionally.

Analysis tools (never given to the optimisers): the reference optimum F*, x_{ML} from 200-start L-BFGS-B, and the Cramér–Rao bound (CRB) J^{-1} with $J = G^T G/\sigma^2$, $G = \partial L^{\text{pred}}/\partial x$, which gives the best achievable unbiased position RMSE and the parameter correlations. Base case: CRB position RMSE 3.4 m, dominated by z (σ_z = 3.2 m vs σ_x = 0.97, σ_y = 0.58 m — the array spans only 10 m vertically); corr(S, α) = 0.94; cond(G) ≈ 340.

Which parameter affects f(x) most? At the best fit, a change of 10 % of each parameter's range raises f by: **S 36.0** dB² > s_y 4.1 > s_x 2.1 > α 0.82 > **s_z 0.02** dB² (per unit: 1 dB of S → +1.0 dB²; 1 m of y/x/z → +0.17 / 0.09 / 0.013 dB²). S shifts every predicted level equally, so ΔS raises f by exactly ΔS². The height changes f about 2,000× less than S, because the microphones span only 10 m vertically while horizontal distances are about 20 m; the weakest curvature direction is almost pure z (component 0.996). Being influential is not the same as being identifiable: S is the most influential parameter on its own, yet it is hard to estimate jointly because α and range compensate it.

3. Method

GWO. The three best positions found so far are the leaders α, β, δ (elitist: best three of old leaders ∪ new pack); every wolf X moves to the average of three leader-guided points,

$$A = 2a \cdot r_1 - a,\ C = 2 \cdot r_2,\ D_p = |C \cdot X_p - X|,\ X'_p = X_p - A \cdot D_p,\ X(t+1) = (X'_\alpha + X'_\beta + X'_\delta)/3$$

with r_1, r_2 ~ U(0,1) per wolf, leader and dimension, and a(t) decreasing from a_0 to 0: linear $a_0(1-\tau)$, nonlinear $a_0(1-\tau^2)$ or exponential $a_0(e^{-5\tau}-e^{-5})/(1-e^{-5})$, τ = t/(T−1). |A| > 1 (only possible while a > 1) throws a wolf beyond its leader — exploration; |A| < 1 lands it between itself and the leader — exploitation. The pack is updated as one vectorised NumPy operation, bounds are reflected after every update and every evaluation is logged. The **ablation** uses $X(t+1) = X'_\alpha$ only.

Baseline: pure random search, uniform in the same encoding box, same objective F, same budget, same per-evaluation log and seeds.

4. Experimental setup

Budget: 9,000 evaluations for every method and setting (30 wolves × [1 + 299] iterations; N = 10 → 899 iterations, N = 60 → 149). **Runs:** 30 per method/setting, run k uses `default_rng(4612 + k)` (common random numbers). **Metrics:** F statistics; localisation error $\|\hat{s} - s\|$; $|\hat{S} - S|$, $|\hat{\alpha} - \alpha|$; success = final estimate within 1 m of x_{ML}; speed = evaluations until best-so-far $F \le F^* + 0.1\sigma^2$, and evaluations after which the incumbent stays within 3 m (≈ base CRB) of the source; Wilcoxon rank-sum tests. **Sensitivity:** pack size {10, 30, 60}, schedule {linear, nonlinear, exponential}, a_0 {1.5, 2.0, 2.5}, on the base and stress scenarios. **Stress:** σ ∈ {1, 3, 5} dB × {20, 16 microphones} on the fixed data set, plus 30 independent noise realisations (one run each). **Shot noise:** random +6…15 dB spikes (probability p per microphone, own seed 4612 + 7000), squared-error vs robust Huber loss, p ∈ {0, 0.1, 0.2, 0.3}. **Location:** 6 source positions × 30 noise draws plus CRB maps of the hall. **Parameter study:** N ∈ {10, 20, 30, 60} × a_0 ∈ {1.5, 2.0, 2.5}. **Extra:** α-only ablation on 3 scenarios and on 10 instances (seeds 4612 + 1000j) × N ∈ {10, 30}; λ_{obs} ∈ {0, 1, 10^2, 10^4}; robustness on 10 further instances.

5. Results

Base (σ = 1 dB, 20 mics), 30 runs	F mean ± sd	median	best	worst	$\|\hat{s}-s\|$ [m]	$\|\hat{S}-S\|$ [dB]	$\|\hat{\alpha}-\alpha\|$ [dB/m]	success	evals → ≤3 m
GWO (N = 30, linear, a_0 = 2)	0.966 ± 0.069	0.954	0.882	1.176	2.01 ± 1.02	1.03	0.035	23 %	2,775 (87 %)
Random search	2.762 ± 0.979	2.521	1.232	5.248	4.71 ± 1.98	1.42	0.055	0 %	5,610 (23 %)
Rank-sum p (GWO vs RS)		$2.9 \cdot 10^{-11}$			$1.1 \cdot 10^{-7}$	0.27	0.038		

F^* = 0.882 (x_{ML} is 0.97 m from the truth). GWO's *worst* run beats random search's *best*: RS samples the 5-D box blindly and almost never hits the narrow low-misfit valley, while GWO concentrates samples around the three best points and shrinks its steps as a → 0. The S error difference is not significant because S is tied to α and range (corr 0.94): near-optimal fits still trade one for the other. Only 23 % of GWO runs end within 1 m of x_{ML} although 67 % reach $F^* + 0.1\sigma^2$ — moving metres along the flat valley floor costs ~0.05 dB^2, below what GWO resolves in 9,000 evaluations; since x_{ML} itself is ~1 m from the truth and the CRB is 3.4 m, this precision limit hardly matters for localisation. 8.4 % of GWO evaluations entered the obstacle and 5.8 % had α < 0, yet no final GWO estimate is infeasible (one RS run ended slightly inside the box).

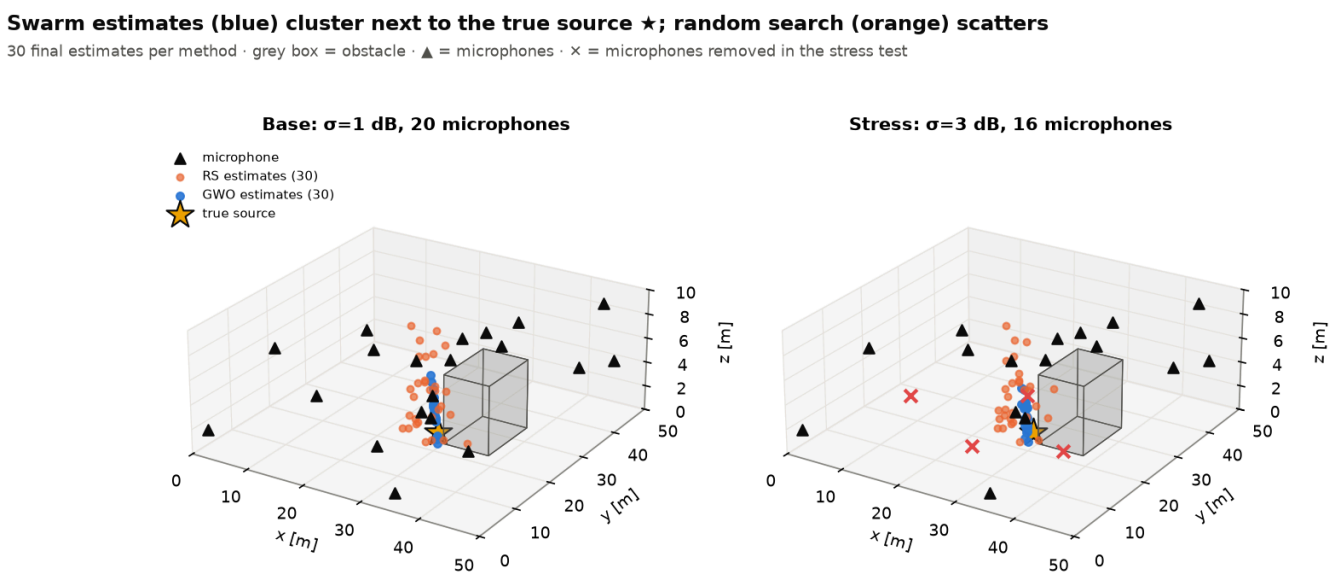


Fig. 1 — Scene: microphones, obstacle, true source (★) and 30 final estimates per method. GWO's estimates spread mainly in z, the least observable coordinate.

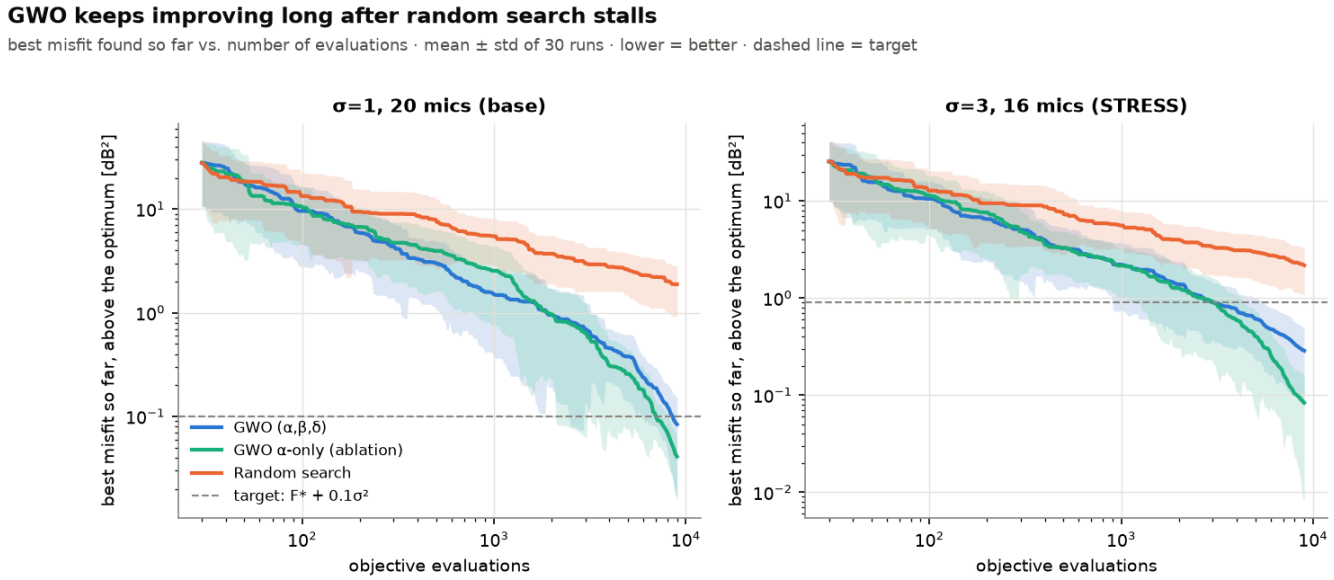


Fig. 2 — Mean ± sd best-so-far F − F* per evaluation (log–log). The methods stay close for the first ~50–100 evaluations, then GWO's information sharing opens a 1–2 order-of-magnitude gap. Under stress the floor is 10× higher and the landscape smoother, so curves flatten earlier.

Parameter sensitivity (30 runs per setting, 9,000 evaluations each)

Setting	N=10	N=30	N=60	linear	nonlinear	exponential	a_0=1.5	a_0=2.0	a_0=2.5
Base: F mean	0.969	0.966	1.015	0.966	0.980	1.003	0.990	0.966	0.951
Base: ‖ŝ−s‖ mean [m]	2.09	2.01	2.78	2.01	2.31	2.32	2.62	2.01	1.80
Base: evals to F target (median)	6,856	7,689	6,811	7,689	8,232	3,221	7,845	7,689	7,594
Stress: success (≤1 m of x_{ML})	43 %	37 %	57 %	37 %	53 %	17 %	7 %	37 %	43 %

At a fixed budget, N = 60 leaves only 149 iterations for the a(t) schedule to refine (worst at σ = 1) but its diversity wins under stress (best success, 57 %). The exponential schedule reaches the F target 2.4× faster but is the least reliable (17 % success under stress): a(t) falls below 1 after ≈ 40 iterations, ending exploration early. a_0 = 1.5 under-explores (|A| > 1 only in the first third of the run); a_0 = 2.5 and the nonlinear schedule, which lengthen the |A| > 1 phase, are the best settings on this flat, coupled landscape (box plots: `fig04/05_*`). **Two-factor study** (N × a_0, hardest scenario σ = 3 dB, 16 mics + shot noise, 30 runs per cell): a_0 = 2.5 is the best column for *every* pack size (excess misfit 0.17–0.21 vs 0.26–0.39 dB², up to 40 % of runs at the best fit), so the interaction is weak; the best cell is N = 20, a_0 = 2.5. The localisation error stays at 5.4–6.2 m in every cell: it is set by the spike-biased optimum, not by the GWO settings (notebook Fig. 17).

6. Stress test

Scenario	F*	GWO F mean	RS F mean	GWO $\|\|\hat{s}-s\|\|$ (fixed data)	GWO RMSE over 30 noise draws	RS RMSE	CRB RMSE	GWO $\|\hat{S}-S\|$ / $\|\hat{\alpha}-\alpha\|$ (draws)
σ=1, 20 mics (base)	0.88	0.97	2.76	2.01 m	3.03 m	5.03 m	3.40 m	0.91 dB / 0.040
σ=3, 20 mics	8.22	8.47	10.62	2.45 m	4.87 m	5.82 m	10.19 m	1.82 / 0.079
σ=1, 16 mics	1.08	1.20	2.79	1.89 m	2.98 m	4.71 m	3.57 m	1.02 / 0.038
σ=3, 16 mics (stress)	10.17	10.46	12.34	2.07 m	4.97 m	6.58 m	10.72 m	2.13 / 0.080
σ=5, 16 mics	29.00	29.37	31.50	2.38 m	6.59 m	7.87 m	17.87 m	3.83 / 0.142

GWO beats RS in every scenario ($p < 10^{-6}$ for F and location). **What degraded and why.** *Noise floor:* F* scales with σ^2, and the CRB position RMSE scales linearly with σ (3.4 → 10.2 m); over 30 noise draws GWO's RMSE grows 3.0 → 5.0 → 6.6 m and the S and α errors roughly quadruple. *Geometry/observability:* removing microphones {0, 14, 18, 19} raises the CRB by only ≈ 5 % but increases corr(S, α) from 0.943 to 0.957 — the array loses mostly the range diversity that separates S from α. *S–α–distance coupling:* on the fixed data set the location barely moves (2.0 → 2.4 m) but the parameters do: at σ ≥ 3 the ML solution pins α at its physical bound 0 and lowers S by 2.5 dB, absorbing the noise by trading attenuation against level (GWO, finding this biased ML solution, has an even larger S error than scattered RS). At σ ≥ 3 the errors fall *below* the CRB because the box ($z \in [0,10]$, $\alpha \geq 0$) caps the estimate: the estimator is biased but its spread is bounded. The optimisation itself does not get harder — GWO hits $F^* + 0.1\sigma^2$ in 100 % of stress runs, median 2,995 evaluations — the *estimation* does. Leader diversity (notebook Fig. 8) stays higher for longer under noise: a flatter valley holds more near-equivalent candidates.

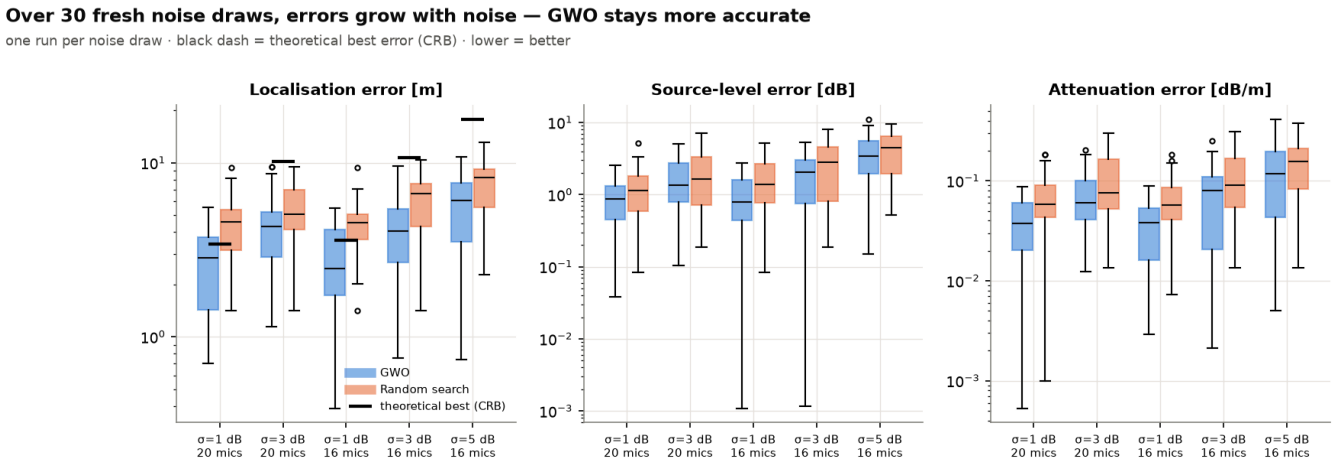


Fig. 3 — Errors over 30 independent noise realisations per scenario (σ / number of microphones); bars mark the CRB RMSE. Noise, not microphone count, drives the degradation.

Shot noise (impulsive spikes)

Short loud events (a door slam, a cough next to a microphone) add *positive* spikes: $L_i^{meas} \mathrel{+}= b_i \cdot u_i$, $b_i \sim$ Bernoulli(p), $u_i \sim U(6, 15)$ dB. With p = 0.15 the seed hits microphones 14 (+10.7 dB) and 17 (+9.6 dB). A 10 dB spike costs 100 dB^2 in the squared error, as much as 100 microphones that are off by 1 dB, so the ML optimum moves **7.7 m** away from the source: a louder reading looks like a closer source. GWO finds that optimum well, but its error vs the truth (6.8 m) is **no better than random search** (6.3 m, p = 0.66): an exact solution of a biased objective is still wrong. A robust **Huber loss** (quadratic up to 2σ, linear beyond) restores 2.8 m (base) and 2.0 m (stress + shot). Over 30 noise draws the squared-error GWO RMSE grows 3.0 → 5.1 → 6.3 → 8.3 m for p = 0 → 0.3; Huber keeps 3.3 → 3.3 → 3.7 → 6.0 m. At p = 0 the squared error is slightly better (it is the true likelihood). At p = 0.3 even random search (7.9 m) beats squared-error GWO.

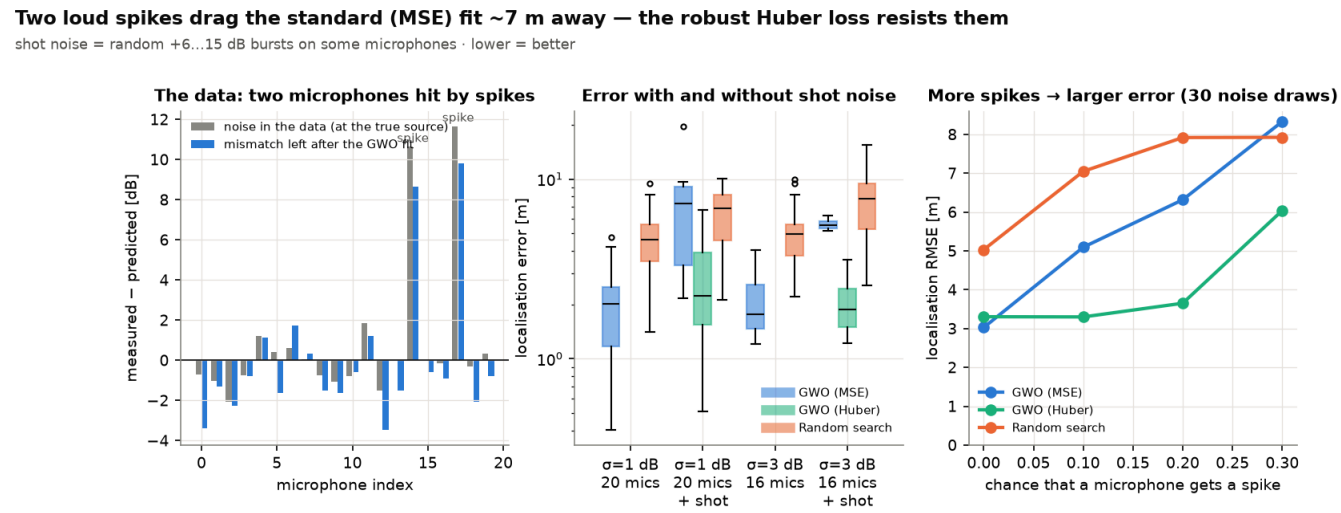


Fig. 4 — Shot noise: the spiked microphones (left), error with and without spikes for squared-error GWO, Huber GWO and random search (middle), and the spike-probability sweep (right).

Where in the hall is the source hard to find?

Same σ = 3 dB and 16 microphones, six source positions, 30 noise draws each. Theory (CRB) says the **centre** of the array is easiest (4.5 m) and **corners** (21.8 m), the area **under the ceiling** (16.7 m) and the wall regions (≈ 11 m) are hardest, because there all microphones see the source from one side. GWO's RMSE ranges from 3.3 m (corner) to **6.7 m (under the ceiling**: the nearest microphone is 12.5 m away and all are below the source). In the corner GWO beats the CRB because the walls and the floor bound the estimate; the CRB ignores this prior information. GWO is significantly better than random search near the obstacle, at the wall and behind the obstacle ($p \leq 0.04$), but not in the centre, corner or under the ceiling.

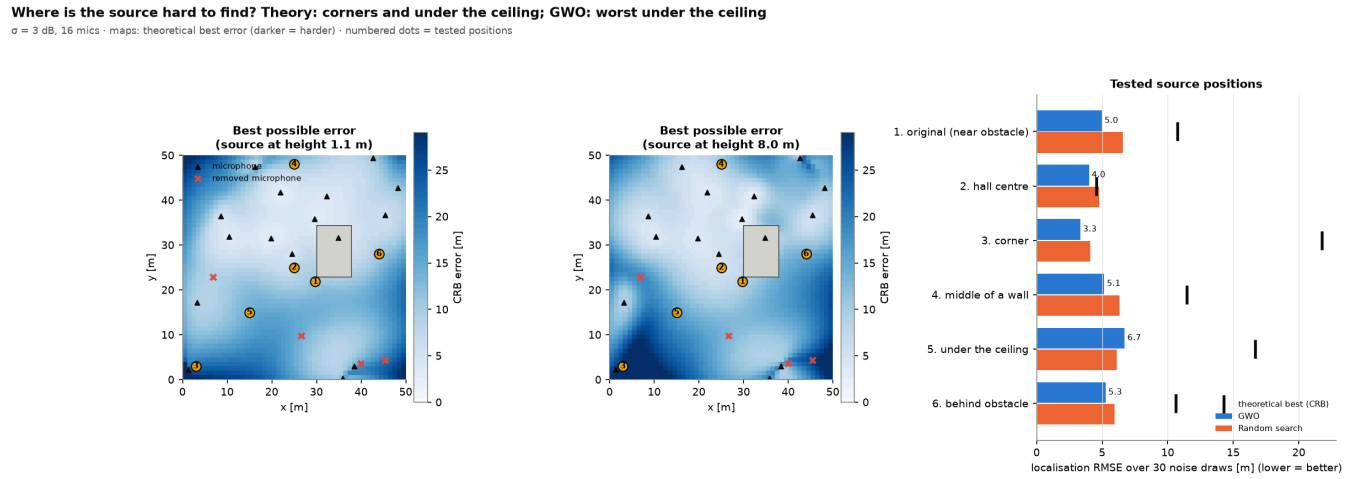


Fig. 5 — Stress by location: maps of the theoretical best error at two source heights (darker = harder; numbered dots = tested positions) and GWO / random-search errors per position.

7. Interpretation: do three leaders protect against one noisy candidate?

Primary instance: no — α-only is better. Localisation 1.36 ± 0.45 vs 2.01 ± 1.02 m (base, p = 0.008), 1.59 ± 0.20 vs 2.07 ± 0.73 m (stress, p = 0.029), 2.07 ± 0.13 vs 2.38 ± 0.41 m (σ = 5, p = 0.0015), with lower spread of final estimates. The three-leader mechanism is visible but does not pay off: averaging three independent moves lowers step variance (mid-run diversity 0.27 vs 0.35) and aims the pack at the centroid of α, β, δ, which on a curved valley lies off the floor; in the second half the three-leader α is > 3 m from x_{ML} 11–20 % of the time vs ≤ 4 % for α-only. α is replaced by a distant "lucky" candidate ≈ 5 times per run in both variants, but on this single-funnel instance those candidates are in the right basin, so greedy following is not punished. Leaders never reach premature consensus: they agree to < 1 m only after ≈ 250 iterations (notebook Fig. 8). **Across 10 instances (σ = 3, 16 mics):** with a small pack (N = 10) three leaders win on 6/10 instances (mean $F - F^*$ 0.84 vs 1.08 dB^2) and avert a catastrophe on instance 6612 (worst run 3.2 vs 24.4 dB^2) — with 10 wolves an α-only pack follows one misleading fit wholesale, while averaging caps any single leader's pull at one third. With N = 30 α-only is better on 7/10 instances and ends far from x_{ML} less often (17 % vs 25 %, signed-rank p = 0.008). **Conclusion:** the three-leader rule is insurance against a single misleading leader that matters when diversity is scarce; when the pack is already diverse it costs exploitation precision.

8. Discussion

Exploration vs exploitation. With linear a(t) and a_0 = 2, half of all moves have |A| > 1 at the start and none after iteration 150 (a = 1; notebook Fig. 7). Leader paths (notebook Fig. 6) show long jumps across the hall in that first phase and short refining moves afterwards; diversity drops sharply after the switch. Schedules that shorten the |A| > 1 phase (exponential, a_0 = 1.5) converge fast but stall; those that lengthen it (nonlinear, a_0 = 2.5) are the most accurate here. **Robustness across seeds and noise.** Over 30 run seeds GWO's F spread is 14× smaller than RS's. On 10 further instances GWO has the lower median error on 9/10 at σ = 1, but only 4/10 under stress: when the CRB is ≈ 7 m the ML estimate itself is metres from the truth, so better fitting no longer means better localising — the limit is information, not optimisation. **Speed.** GWO's incumbent stays within 3 m of the source after a median 2,775 evaluations (87 % of runs) vs 5,610 for RS (23 %); the F target is reached after 7,689 evaluations at σ = 1 and 2,995 under stress. **Failure modes** (fig15, failure table): (i) *stagnation when a(t) decays too fast* — exponential decay leaves 77 % of stress runs short of F* (57 % linear); (ii) *obstacle-boundary solution* — on instance 5612 the ML optimum lies on the obstacle roof (z = 4.49 m, top 4.5 m), 7 m from the source, and runs split between two faces of the box; (iii) *S–α–distance ambiguity* — on instance 13612 x_{ML} is 6.6 m from the truth with $\hat{\alpha}$ = 0.32 vs 0.11 dB/m: a closer, more strongly attenuated source explains the data equally well, and 43 % of runs stop in the flat valley between. (ii) and (iii) are properties of the data: a better optimiser converges more reliably to the wrong-but-optimal answer. **λ check:** $\lambda_{obs} \in \{0, 1, 10^2, 10^4\}$ leaves no estimate inside the box on instance 4612 (constraint inactive at the optimum); 10^4 is slightly worse (2.11 vs 1.9 m) because a steep wall makes the region next to the box harder to explore.

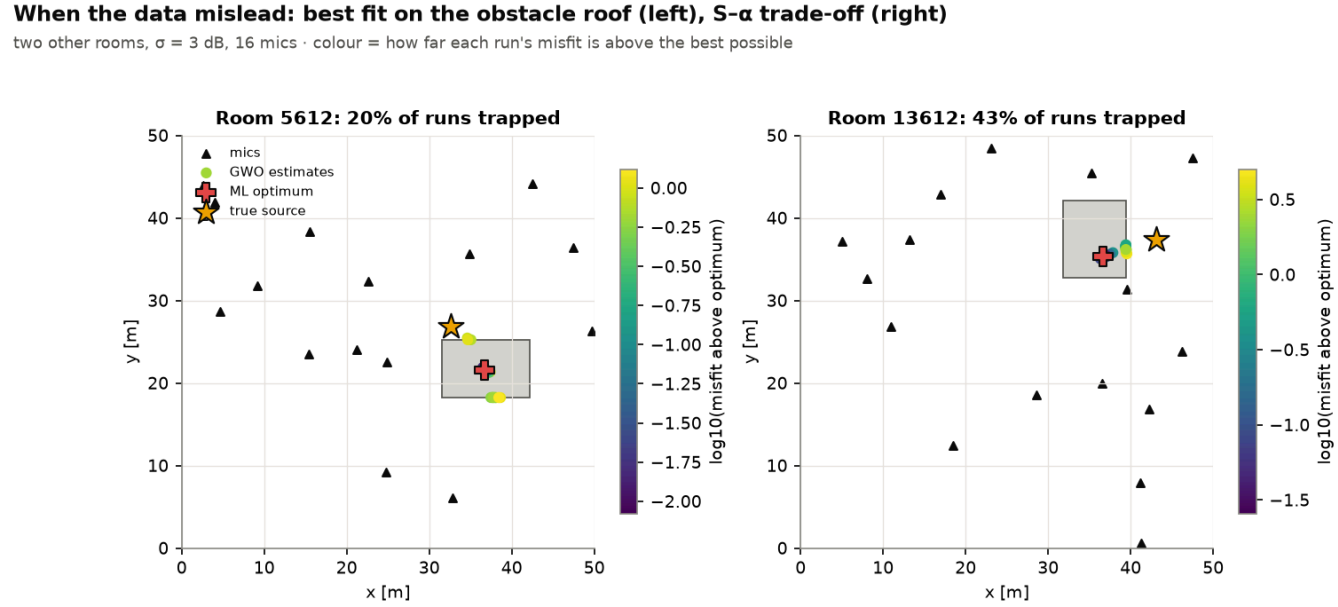


Fig. 6 — Failure modes: obstacle-boundary ML solution (instance 5612) and S–α–distance ambiguity (instance 13612); colour = $\log_{10}(F - F^*)$ of each run's final estimate.

9. Conclusion

A from-scratch GWO with reflection bounds and a graded depth penalty solves the 5-parameter localisation problem far better than random search at the same 9,000-evaluation budget (F 0.97 vs 2.76 dB^2, error 2.0 vs 4.7 m, $p \le 10^{-7}$) and reaches near-CRB accuracy at σ = 1. Its accuracy is limited by the physics, not by the search. The height z changes the misfit about 2,000× less than S, and S, α and distance compensate each other. Under 3–5 dB noise any estimator degrades. Accuracy also depends on where the source is: best in the centre of the array, worst under the ceiling and outside the array. Impulsive shot noise is the most damaging stressor: two spikes move the squared-error optimum 7.7 m, and only a robust (Huber) objective recovers 2–3 m accuracy. With a biased objective, a better optimiser just finds the wrong answer more reliably. Longer exploration (a_0 = 2.5, nonlinear a(t)) helps for every pack size; fast decay hurts. The three-leader guidance is robustness insurance for small packs rather than a universal gain. **Recommendation:** GWO with $a_0 \approx 2.5$ and 20–30 wolves, a Huber misfit whenever spikes are possible, and microphones that surround the expected source region at several heights. Natural extensions are a final local polish to resolve the flat valley and a reparameterisation (e.g. $S - \alpha\cdot\bar{r}$) to decorrelate S and α.